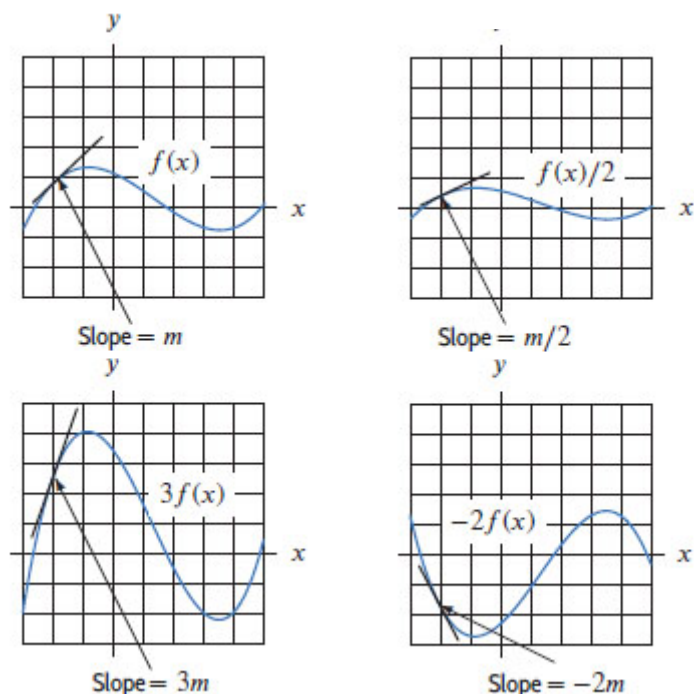


Sec 3.1 – Powers and Polynomials

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Below the function $y = f(x)$ is graphed, as well as three multiples of this function, $3f(x)$, $\frac{f(x)}{2}$, $-2f(x)$. What is the relationship of the derivatives of these functions? In other words, for a particular x value, how are the slopes of the tangent lines of these functions related?



Multiplying the value of a function by a constant stretches or shrinks the original graph (and reflects it across the x -axis if the constant is negative). This changes the slope of the curve at each point.

If the graph has been stretched, the “rises” have all been increased by the same factor, whereas the “runs” remain the same. Thus, the slopes are all steeper by the same factor.

If the graph has been shrunk, the slopes are all smaller by the same factor.

If the graph has been reflected across the x -axis, the slopes will all have their signs reversed.

In other words, if a function is multiplied by a constant, c , so is its derivative:

Let $F(x) = cf(x)$ for some constant c .

Then

$$\begin{aligned}
 F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{c(f(x+h) - f(x))}{h} \\
 &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ by the Constant Multiple Law for Limits} \\
 &= cf'(x).
 \end{aligned}$$

Similarly, if we add or subtract differentiable functions, we can take the derivatives of the separate functions:

Sum and Difference Rules: $f + g$ and $f - g$ are differentiable and

$$(f(x) + g(x))' = f'(x) + g'(x) \quad (f(x) - g(x))' = f'(x) - g'(x)$$

$$(Alternately, \frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x))$$

In Chap 2 we presented the Power Rule, and for the rest of the semester it will be used and applied just about every day!

Theorem (The Power Rule)

For any constant real number n ,

$$\frac{d}{dx}x^n = nx^{n-1}$$

In words, we would say that the derivative of x to an exponent is that exponent times x to one less than that exponent. You might remember this as “bring down the exponent and subtract 1 from the exponent.”

Partner up

Find each of the following derivatives:

① $\frac{d}{dx}(x^3 + 4)$

② $\frac{d}{dt}(t^3 - 3t + \pi)$

③ $\frac{d}{dx}(\sqrt{x} + 4x)$

① $\frac{d}{dx}(x^3 + 4) = 3x^2 + 0 = 3x^2$

② $\frac{d}{dt}(t^3 - 3t + \pi) = 3t^2 - 3 + 0 = 3t^2 - 3$

③ $\frac{d}{dx}(\sqrt{x} + 4x) = \frac{d}{dx}(x^{1/2} + 4x) = \frac{1}{2}x^{-1/2} + 4$

Find each of the following derivatives:

① $\frac{d}{dx} (2x^4 - 3x^2 + 5)$

② $\frac{d}{dx} x^{\sqrt{2}}$

③ $\frac{d}{dx} \frac{1}{x^2}$

④ $\frac{d}{dt} (\sqrt{2}t^2 - 3t^{-1})$

⑤ $\frac{d}{dx} \left(x^{3/2} + \frac{7}{\sqrt{x}} \right)$

⑥ $\frac{d}{dx} \frac{x^3 - 5x + 2}{x^2}$

⑦ $\frac{d}{dt} (at^3 + bt + c)$, where a, b, c are constants.

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① $\frac{d}{dx} (2x^4 - 3x^2 + 5) = 8x^3 - 6x$

② $\frac{d}{dx} x^{\sqrt{2}} = \sqrt{2}x^{\sqrt{2}-1}$

③ $\frac{d}{dx} \frac{1}{x^2} = \frac{d}{dx} x^{-2} = -2x^{-3} = \frac{-2}{x^3}$

④ $\frac{d}{dt} (\sqrt{2}t^2 - 3t^{-1}) = 2\sqrt{2}t + 3t^{-2}$

⑤ $\frac{d}{dx} \left(x^{3/2} + \frac{7}{\sqrt{x}} \right) = \frac{d}{dx} \left(x^{3/2} + 7x^{-1/2} \right) = \frac{3}{2}x^{1/2} - \frac{7}{2}x^{-3/2}$

⑥ $\frac{d}{dx} \frac{x^3 - 5x + 2}{x^2} = \frac{d}{dx} (x - 5x^{-1} + 2x^{-2}) = 1 + \frac{5}{x^2} - \frac{4}{x^3}$ Do the division first!

⑦ $\frac{d}{dt} (at^3 + bt + c) = 3at^2 + b$ (Treat the constants as numbers.)

Find all tangent lines to $f(x) = x^3 - 7x^2 - 3x + 4$ that have slope 2.

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Solution: To solve this, we need to find the x -values where the slope of the tangent line is 2; that is, we need to know where $f'(x) = 2$. Since $f'(x) = 3x^2 - 14x - 3$, $f'(x) = 2$ when $3x^2 - 14x - 3 = 2$. Therefore, $3x^2 - 14x - 5 = 0$ and factoring, we get $(3x + 1)(x - 5) = 0$, so $x = -1/3$ or $x = 5$.

We find a tangent for each x -value. Since $f(-1/3) = 113/27$, the tangent line at $x = -1/3$ is $y = 2(x + 1/3) + 113/27$. Since $f(5) = -61$, the tangent line at $x = 5$ is $y = 2(x - 5) - 61$.

Consider the function $g(x) = x^3 - 12x$. On which intervals is it decreasing? On which intervals is it concave down?

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Solution: To determine where the function is decreasing, we need to know where $g'(x) < 0$. Since $g'(x) = 3x^2 - 12$, we need to know where $3x^2 - 12 < 0$. We switch to an equation to simplify and then we will find sections that are above or below 0. When $3x^2 - 12 = 0$, we have $3(x - 2)(x + 2) = 0$, which means $x = 2$ or $x = -2$. This splits x -values into the intervals $(-\infty, -2)$, $(-2, 2)$ and $(2, \infty)$. We check a value in each to realize that $g'(x) < 0$ on the interval $(-2, 2)$.

To determine where the function is concave down, we need to know where $g''(x) < 0$. This occurs when $g''(x) = 6x < 0$, which is exactly when $x < 0$. Therefore the function is concave down on $(-\infty, 0)$.